605.486 – Android Development

Module 29
Wifi

Note!

- If you're running an emulator and trying to reach a server on the host machine…
- 10.0.2.2 is localhost/127.0.0.1 for the host

Wifi - Server

- Uses standard Java Socket Communication
- Standard Java Application
  - Determine and print IP Address
  - Thread accepts socket connections from client
  - Passes handling to a separate handler
  - Receive thread
    - Gets requests from client
    - Queues response on send thread
  - Send thread
    - Sends queued responses to client
    - Allows other actions on server to communicate with client
      - Not always needed – can often send response on same thread client sent request
  - Cleanup
    - Close all streams and sockets

Wifi – Client (1)

- Android application
- Open socket to server
- Set timeout (IO blocks – timeout allows us to close)
- Send request by writing to socket's output stream
- Receive response on thread (needed if server sends on thread)

Wifi – Client (2)

- Cleanup – close all streams and socket
- Need INTERNET permission
- Caution – cannot communicate on GUI thread!